Fiscal Year:	FY 2016	**Task Last Updated:**	FY 08/02/2016
PI Name:	Crucian, Brian Ph.D.		
Project Title:	Deep Space Exploration Atmosphere Effects on Hematologic, Immunologic, and Oxidative Stress and Damage Parameters in Astronauts: An ISS Flight Study		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **EVA**:Risk of Mission Impacting Injury and Compromised Performance and Long-Term Health Effects due to EVA Operations (2) **Immune**:Risk of In Mission Impacts, Adverse Health Events or Long-Term Health Impacts due to Altered Immune Response		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058-3607	**Congressional District:**	36
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2013 HERO NNJ13ZSA002N-Crew Health (FLAGSHIP & NSBRI)
Start Date:	10/01/2014	**End Date:**	07/15/2016
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Norsk, Peter	**Contact Phone:**	
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 7/15/2016 due to project cancellation (original end date was 9/30/2019), per JSC HRP (Ed., 8/9/16)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Goodwin, Thomas Ph.D. (NASA Johnson Space Center) Basner, Mathias (University of Pennsylvania) Otto, Christian (Universities Space Research Association) Simpson, Richard (University of Houston) Chouker, Alexander (ESA/University of Munich)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	AUGUST 2016 FINAL REPORTING: “HIOS” (Hematologic, Immunologic, and Oxidative Stress) was originally selected as a flight study to examine the combined effects of spaceflight and hypobaric hypoxia on the immune system; however, due to operational and crew time constraints, the study was cancelled in July of 2016. Despite the cancellation of the study, various procedures and protocols were developed during the flight definition year of this study, including on-orbit culture systems and methods for long-term sample preservation. The procedures developed in support of this study are relevant for future in-flight studies or for remote immune monitoring and have the potential to expand our science capabilities on the International Space Station (ISS). As originally proposed, this study aimed to characterize and/or validate the combined effects of spaceflight and the persistent hypobaric hypoxia of the exploration atmosphere on the immune system. Spaceflight-adaptation syndrome is the combination of space environmental conditions and the synergistic reaction of the human physiology. Onboard the ISS, astronauts currently live in an atmosphere environment similar to Earth (21% Oxygen; 14.7 psi pressure). Recent studies have demonstrated that astronauts have significant alterations to their immune system that persist for the duration of a 6-month spaceflight. During longer exploration missions, these immune changes may increase astronaut health risks. On Earth, it is well established that hypoxia can influence the immune system, as well as increase markers of oxidative damage. NASA has decided on an atmosphere for deep space exploration missions that is considered hypoxic (34% O2; 8.2 psi pressure). This study, therefore, sought to characterize the synergistic effects of spaceflight and the proposed “exploration atmosphere” on the immune system. Our proposal suggested to expose ISS crewmembers to the exploration atmosphere during flight, using either overnight stays in the US Airlock module or portable breathing apparatus. We have previously examined immunity during spaceflight onboard ISS. Blood and saliva samples were collected on-orbit, and immediately returned for analysis. For this proposal, it would not have been feasible to have the atmosphere exposure occur near to a vehicle undock. Therefore, a unique suite of immune and oxidative sampling and processing that are compatible with sample storage onboard ISS for several weeks were developed. These procedures allow for exposure of ISS crewmembers to the exploration atmosphere during spaceflight, and still completely characterize their immune and oxidative stress responses. As no terrestrial space analog can completely recreate the influences of spaceflight (microgravity, radiation, stress, disrupted circadian rhythms), it was important that this validation be a flight study onboard ISS. Only on ISS is it possible to look at the synergistic effect of persistent hypobaric hypoxia, in astronauts already experiencing the physiological dysregulation of spaceflight. We proposed to collect samples before flight to establish a true subject baseline, during flight several days before atmosphere exposure to establish “space normal” for each subject, and following atmosphere exposure to establish the influence of the exploration atmosphere. A post-EVA (extravehicular activity) sample was requested, which would have allowed a similar assessment of transient hyperoxia. For this study, it was proposed that saliva and blood plasma samples would be immediately frozen for storage and subsequent return to Earth. For cellular function analyses, procedures for setting up and processing whole blood cultures on orbit were developed such that cells and purified culture supernatants could be frozen for analysis upon return to Earth. Additional blood samples were to be collected into special tubes, one containing a unique cell preservative that maintains cell integrity for up to several weeks. This tube would have been used to count and characterize in-vivo leukocyte distribution. The other tube contains a chemical that lyses cells but preserves RNA and DNA, and would have been used for various molecular analyses. This study sought to address the role of oxidative stress and damage (OSaD) as a negative and contributing Risk Factor (RF) in the following areas of combined spaceflight related dysregulation: i) immune impacts and the inflammatory response (Bagai et al., 2009; Crucian et al., 2008, 2013); and ii) Altered Exploration atmospheres proposed for EVA and IVA exploration missions (Schmidt and Goodwin 2013). Although the study was cancelled, the unique sample collection and preservation techniques that were developed in support of this study may be used for future studies, enhancing the science capabilities onboard ISS.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	AUGUST 2016 FINAL REPORTING: Data from the originally proposed study may have proven useful for understanding the physiology and disease susceptibility for populations living at altitude. Terrestrial data regarding hypoxia and immunity may vary depending on the population and degree of hypoxia/hypobaria. This study would have provided new insights into the effect of hypobaric hypoxia on a variety of immune, hematologic, and oxidative stress parameters. While the proposed study was not performed, the preservation techniques developed for this project expand immune monitoring capabilities both on the ISS and in remote locations. With the current capabilities, immune assessment is limited to frozen saliva, urine, and plasma, or is tied to vehicle undockings, during which ambient samples are returned to Earth. The stressors associated with undocking may alter the immune parameters of interest, and therefore data from these time points may not be the most representative samples of the in-flight condition. The hardware and long-term sample storage procedures developed for this study would permit sampling throughout an ISS mission and eliminate the need to coordinate sampling with vehicle undockings. JULY 2015 ANNUAL REPORTING: The Terrestrial data regarding hypoxia and immunity may vary depending on the population and degree of hypoxia/hypobaria. This study will provide new insights into the effect of hypobaric hypoxia on a variety of immune, hematologic, and oxidative stress parameters. This data may prove useful for understanding the physiology and disease susceptibility for populations living at altitude.
Task Progress:	AUGUST 2016 FINAL REPORTING: Project was Cancelled in July 2016; Sampling and preservation techniques were developed in preparation for the study that can be utilized for future work onboard ISS or in remote locations. A manuscript detailing the unique preservation and storage techniques is currently in preparation. JULY 2015 ANNUAL REPORTING: Major activities for this study were placed on hold status during 2015, as NASA has engaged a panel of experts to evaluate the research plan to evaluate/validate the exploration atmosphere for flight. This includes plans for ground validation of the atmosphere with healthy test subjects and flight validation onboard ISS. Pending the findings of this review, planned for August 2015, the current flight study may go forward, be modified, or deselected for flight.

Bibliography Type:	Description: (Last Updated: 05/21/2026)
Abstracts for Journals and Proceedings	Crucian B, Chouker A, Pierson D, Simpson R, Goodwin T. "The ‘Exploration Atmosphere’ Flight Study on ISS: Assessment of Immunity, Oxidative Stress, Vision and Cognition." 2015 NASA Human Research Program Investigators’ Workshop, Galveston, TX, January 13-15, 2015. 2015 NASA Human Research Program Investigators’ Workshop, Galveston, TX, January 13-15, 2015 , Jan-2015